

Activity-1

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BSCS - 3A

CS311
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Code

```
1  using System;
2  using Avalonia.Controls;
3  using Avalonia.Interactivity;
4
5  namespace frmPrelimActivity1;
6
7  public class CleanException : System.Exception
8  {
9      public CleanException(string message) : base(message) { }
10
11     // Overriding this removes the "System.Exception:" prefix when printing the
12     // exception
13     public override string ToString()
14     {
15         return Message;
16     }
17
18     public partial class MainWindow : Window
19     {
20         public MainWindow()
21         {
22             InitializeComponent();
23         }
24
25         private bool parseOpposite(out double opposite)
26         {
27             DataValidationErrors.ClearErrors(OppositeInput);
28             if (String.IsNullOrEmpty(OppositeInput.Text))
29             {
30                 DataValidationErrors.SetError(OppositeInput,
31                     new CleanException("Opposite should not be empty."));
32                 opposite = 0;
33                 return false;
34             }
35
36             if (!double.TryParse(OppositeInput.Text, out opposite))
37             {
38                 DataValidationErrors.SetError(OppositeInput,
```

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39         new CleanException("Opposite should be a valid number.));
40         return false;
41     }
42
43     return true;
44 }
45
46 private bool parseHypotenuse(out double hypotenuse)
47 {
48     DataValidationErrors.ClearErrors(HypotenuseInput);
49     if (String.IsNullOrEmpty(HypotenuseInput.Text))
50     {
51         DataValidationErrors.SetError(HypotenuseInput,
52             new CleanException("Hypotenuse should not be empty.));
53         hypotenuse = 0;
54         return false;
55     }
56
57     if (!double.TryParse(HypotenuseInput.Text, out hypotenuse))
58     {
59         DataValidationErrors.SetError(HypotenuseInput,
60             new CleanException("Hypotenuse should be a valid number.));
61         return false;
62     }
63
64     return true;
65 }
66
67 private bool validateAdjacent(out double adjacent)
68 {
69     DataValidationErrors.ClearErrors(AdjacentInput);
70     if (String.IsNullOrEmpty(AdjacentInput.Text))
71     {
72         DataValidationErrors.SetError(AdjacentInput,
73             new CleanException("Adjacent should not be empty.));
74         adjacent = 0;
75         return false;
76     }
77
78     if (!double.TryParse(AdjacentInput.Text, out adjacent))
79     {
80         DataValidationErrors.SetError(AdjacentInput,
81             new CleanException("Adjacent should be a valid number.));
82         return false;
83     }
84
85     return true;
```

```

86     }
87
88     private void ComputeSineButton_OnClick(object? sender, RoutedEventArgs e)
89     {
90
91         DataValidationErrors.ClearErrors(AdjacentInput);
92         // singular & to disable short-circuiting
93         if (parseOpposite(out double opposite) & parseHypotenuse(out double
hypotenuse))
94         {
95             double sine = opposite / hypotenuse;
96             SineOutput.Text = sine.ToString("0.00");
97         }
98     }
99
100    private void ComputeCosineButton_OnClick(object? sender, RoutedEventArgs e)
101    {
102        DataValidationErrors.ClearErrors(OppositeInput);
103        if (validateAdjacent(out double adjacent) & parseHypotenuse(out double
hypotenuse))
104        {
105            double cosine = adjacent / hypotenuse;
106            CosineOutput.Text = cosine.ToString("0.00");
107        }
108    }
109
110    private void ComputeTangentButton_OnClick(object? sender, RoutedEventArgs
e)
111    {
112        DataValidationErrors.ClearErrors(HypotenuseInput);
113        if (parseOpposite(out double opposite) & validateAdjacent(out double
adjacent))
114        {
115            double tangent = opposite / adjacent;
116            TangentOutput.Text = tangent.ToString("0.00");
117        }
118    }
119
120    private void ClearAllButton_OnClick(object? sender, RoutedEventArgs e)
121    {
122        DataValidationErrors.ClearErrors(OppositeInput);
123        DataValidationErrors.ClearErrors(HypotenuseInput);
124        DataValidationErrors.ClearErrors(AdjacentInput);
125        OppositeInput.Text = "";
126        HypotenuseInput.Text = "";
127        AdjacentInput.Text = "";
128        SineOutput.Text = "";

```

```

129     CosineOutput.Text = "";
130     TangentOutput.Text = "";
131 }
132
133 private void ClearButton_OnClick(object? sender, RoutedEventArgs e)
134 {
135     OppositeInput.Text = "";
136     HypotenuseInput.Text = "";
137     AdjacentInput.Text = "";
138     DataValidationErrors.ClearErrors(OppositeInput);
139     DataValidationErrors.ClearErrors(HypotenuseInput);
140     DataValidationErrors.ClearErrors(AdjacentInput);
141     SineOutput.Text = "";
142     CosineOutput.Text = "";
143     TangentOutput.Text = "";
144 }
145
146 private void ComputeAllButton_OnClick(object? sender, RoutedEventArgs e)
147 {
148     if (parseOpposite(out double opposite) & parseHypotenuse(out double
hypotenuse) & validateAdjacent(out double adjacent))
149     {
150         double sine = opposite / hypotenuse;
151         SineOutput.Text = sine.ToString("0.00");
152         double cosine = adjacent / hypotenuse;
153         CosineOutput.Text = cosine.ToString("0.00");
154         double tangent = opposite / adjacent;
155         TangentOutput.Text = tangent.ToString("0.00");
156     }
157 }
158 }

```

Screenshots

1. Opposite and Hypotenuse are empty and click sine button.

Adjacent

Hypotenuse

Hypotenuse should not be empty.

Opposite

Opposite should not be empty.

Sine

Cosine

Tangent

2. Adjacent and Hypotenuse are empty and click cosine button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Hypotenuse should not be empty.

Opposite

Sine

Cosine

Tangent

3. Opposite and Adjacent are empty and click tangent button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Opposite

Opposite should not be empty.

Sine

Cosine

Tangent

4. All inputs are empty and click compute all button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Hypotenuse should not be empty.

Opposite

Opposite should not be empty.

Sine

Cosine

Tangent

5. Adjacent is not a number.

Adjacent
Adjacent should be a valid number.

Hypotenuse
Hypotenuse should not be empty.

Opposite

Sine

Cosine

Tangent

6. Hypotenuse is not a number.

Adjacent

Hypotenuse
Hypotenuse should be a valid number.

Opposite
Opposite should not be empty.

Sine

Cosine

Tangent

7. Opposite is not a number.

Adjacent

Adjacent should not be empty.

Hypotenuse

Opposite

Opposite should be a valid number.

Sine

Cosine

Tangent

8. Hypotenuse is 3 and Opposite is 6 and click sine button.

Adjacent

Hypotenuse

Opposite

Sine

Cosine

Tangent

9. Hypotenuse is 6 and Opposite is 3 and click sine button. Result should be in 2 decimal places only.

Adjacent	
Hypotenuse	6
Opposite	3
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	0.50
Cosine	
Tangent	

10. Adjacent is 10 and Hypotenuse is 5 and click cosine button.

Adjacent	10
Hypotenuse	5
Opposite	
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	
Cosine	2.00
Tangent	

11. Adjacent is 5 and Hypotenuse is 10 and click cosine button. Result should be in 2 decimal places only.

Adjacent	5
Hypotenuse	10
Opposite	
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	
Cosine	0.50
Tangent	

12. Adjacent is 8 and Opposite is 5 and click tangent button. Result should be in 2 decimal places only.

Adjacent	8
Hypotenuse	
Opposite	5
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	
Cosine	0.50
Tangent	0.63

13. Adjacent is 8, Hypotenuse is 5, and Opposite is 3 and click compute all button.

Adjacent	8
Hypotenuse	5
Opposite	3
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	0.60
Cosine	1.60
Tangent	0.38

14. Adjacent is 3, Hypotenuse is 5, and Opposite is 8 and click compute all button.

Adjacent	3
Hypotenuse	5
Opposite	8
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	1.60
Cosine	0.60
Tangent	2.67

15. Adjacent is 5, Hypotenuse is 8, and Opposite is 3 and click compute all button. Result should be in 2 decimal places only

Adjacent	5
Hypotenuse	8
Opposite	3
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	0.38
Cosine	0.63
Tangent	0.60

<https://drive.google.com/file/d/1w2RrUu6AvIlk6oHzqqRsgssAXj9rJIsc/view?usp=sharing>